

Indian Institute of Technology, Gandhinagar

EE226 Semiconductor Devices

Semester 2, 2025-2026

Tutorial - 3

Q1. Is an electron a particle or a wave? Discuss any one experiment which can be conducted to find electron nature.

Q2. Describe E-k relationship for a free electron and electron in a crystal.

- (a) Which quantity is used to incorporate all the quantum/wave nature and describe electron as if free in a crystal?
- (b) What information does the double slope of E-k relationship provide?

Q3. What is the density of states (DOS) i.e $g(E)$? What is the relation of DOS with Energy? Plot E vs $g(E)$. Also, plot $g(E)$ vs E for visualization in different ways.

Q4. Why does an electron have an occupancy probability ($f(E)$)?

- (a) Plot E vs $f(E)$ and also plot $f(E)$ vs E for visualization in different ways.
- (b) What is the Fermi level?
- (c) How does probability change with temperature? Describe the probability below and above Fermi level energy for $T_1=0K$, $T_2>T_1$, and $T_3>T_2$

Q5. How to find carrier concentration $n(E)$ or $p(E)$? How to find total carrier concentration “n” or “p”?

Q6. Answer the following:

- (a) What is the intrinsic carrier concentration at 300K?
- (b) 300K would lead to thermal energy (kT) of $\sim 25meV$ whereas to generate an electron-hole pair, at least energy equal to the bandgap (1.1eV for Si) would be required. How come we get intrinsic carrier concentration?
- (c) Can a few electrons, each with 26meV energy pass their energy to excite one electron from VB to CB?

Q7. Graphically show the band diagram, DOS, occupancy probability, and carrier concentration for an intrinsic semiconductor.

- (a) Why are there no carriers between E_c and E_v ?
- (b) Where are the maximum number of carriers located? Are they exactly at E_c or E_v ?

Q8. Plot the intrinsic carrier concentration for Si with Temperature. Describe the dependence.

Q9. Plot the intrinsic carrier concentration with bandgap varying from 0.5eV to 3eV, assume $N_c=N_v=10^{19}/cm^3$ and $T=300K$

Q10. Describe the alternate expression for intrinsic concentration. How does it help?

Q11. Where is E_i located for an intrinsic semiconductor?

- (a) Is it at the centre of the bandgap?
- (b) If not at the centre, then $E_i = E_f$ will affect the occupancy probability, which should give different electron and hole concentrations. Why don't we see different electron-hole concentrations for intrinsic semiconductor?

Q12. For a given semiconductor, the fermi level lies at 6.21 eV. Find the probability of occupancy of an energy level with energy 6.25 eV at temperatures of 300K and 370K. Given $k_B = 8.62 \times 10^{-5}$ eV/K.

Q13. Based on the results obtained in Question 12, comment in which temperature the probability of occupancy is high? Using the plot of Fermi Dirac distribution function, graphically show that your calculations are matching the trend.

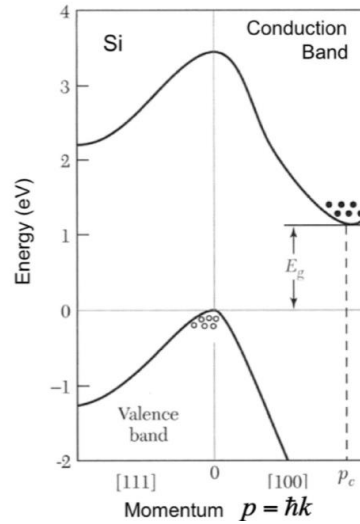
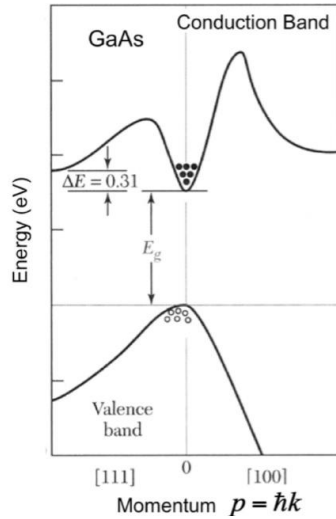
Q14. A hypothetical semiconductor has an intrinsic carrier concentration of $1.0 \times 10^{10} / \text{cm}^3$

at 300 K, it has conduction and valence band effective densities of states N_c and N_v both equal to 10^{19} cm^3

- (a) What is the bandgap?
- (b) If the semiconductor is doped with $N_d = 1 \times 10^{16}$ donors/ cm^3 , what are the equilibrium electron and hole concentrations?
- (c) If the same piece of semiconductor, already having $N_d = 1 \times 10^{16}$ donors/ cm^3 , is also doped with $N_a = 2 \times 10^{16}$ acceptors/ cm^3 , what are the new equilibrium electron and hole concentrations at 300 K?
- (d) Consistent with your answer to part (c), what is the Fermi level position with respect to the intrinsic Fermi level, $E_f - E_i$? For $E_i - E_f = 0.3$ eV what would be acceptor doping?
- (e) Draw the equilibrium energy band diagram for $N_a = 1 \times 10^{17}$ donors/ cm^3 . What are the equilibrium electron and hole concentrations?
- (f) Draw the equilibrium energy band diagram for $N_d = 1 \times 10^{18}$ donors/ cm^3 . What are the equilibrium electron and hole concentrations?

Q15. Why is the intrinsic carrier concentration n_i strongly dependent on the bandgap E_g ? Plot E_g vs n_i

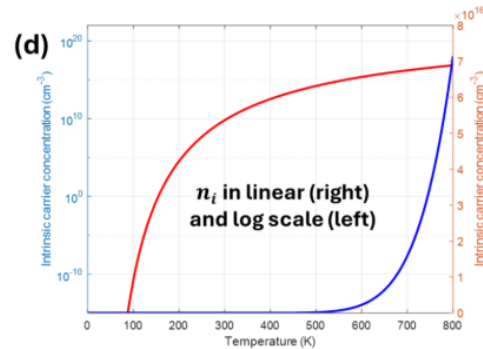
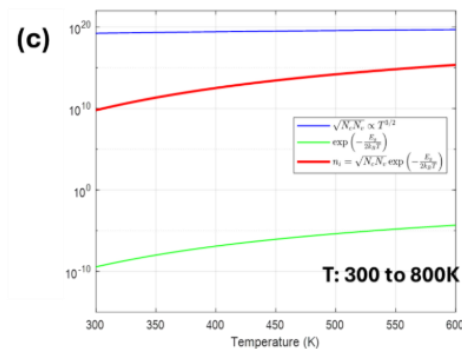
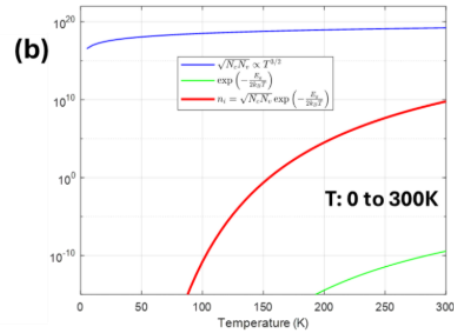
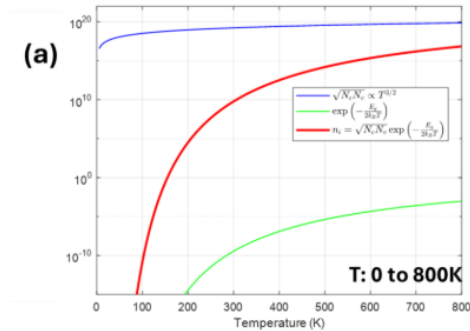
Q16. The figure below shows the E-k diagrams for GaAs and Si. Which material appears to have the lowest electron effective mass in the conduction band? Also define the regions where electrons and holes have positive and negative effective mass. What does negative mass mean?



Q17. For intrinsic carrier concentration

- Describe various temperature dependencies
- See the figures below (plotted in different T ranges) (a) T: 0 - 800K, (b) T: 0 - 300K, (c) T: 300 - 800K ; which temperature dependency is more dominant in which temperature range?
- Also, observe the n_i in linear and log scales.

Throughout the course, you will notice that there will be multiple types of dependencies of various parameters (e.g. here different temperature dependence for n_i). Try to see which dependence dominates more than the others. MATLAB Codes shared in resources folder for practise and visualization .



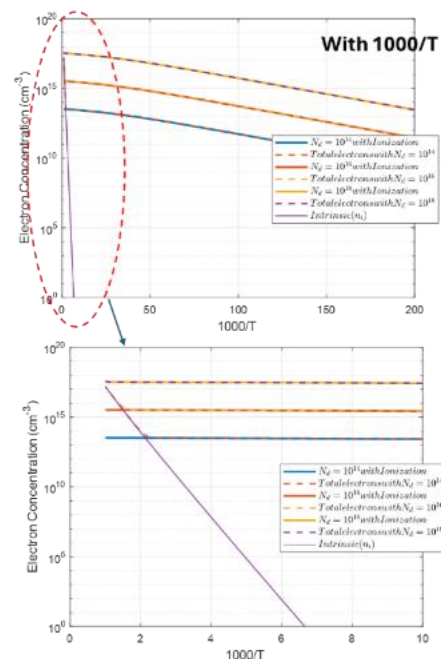
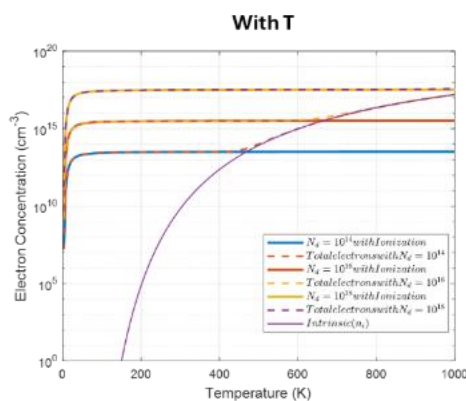
Q18. How can the Fermi level (E_F) can be thought of as a separation between electron and hole concentrations ($\#/cm^3$). Describe using the mass action law $n_0.p_0 = n_i^2$ Describe the electrons and holes concentration ($\#/cm^3$) with the Fermi level moving towards EC and EV .

Q19. Read section 2.13 (Degenerate Semiconductor) from the book “Fundamentals of Semiconductor Devices” by B. Anderson & R. Anderson. Answer the following questions.

- Will the bandgap remain the same as compared to the intrinsic semiconductor bandgap? Why?
- Is mass action law ($n_0.p_0 = n_i^2$) valid for degenerate semiconductors? Justify.
- What will be $n_0.p_0$ product for a degenerate semiconductor?

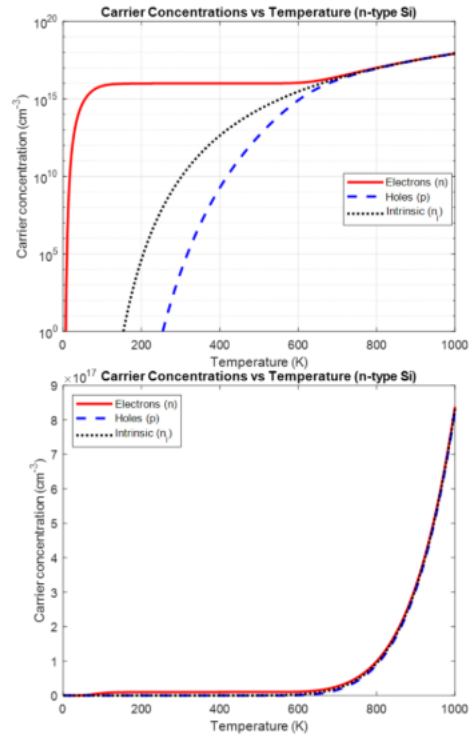
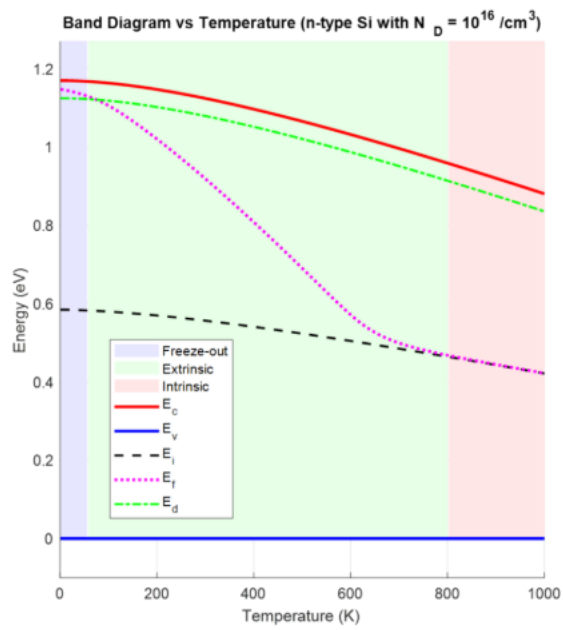
Q20. Read the section 3.3.4 Compensation and Space Charge neutrality from Streetman book.

- Describe Ionization with temperature.
- How to find the electron and hole concentration in different temperature ranges.
- See the figure below and how to plots look with T , $1/T$: Codes shared in resources folder for practise and visualization.
 - Find the temperature needed for an extrinsic semiconductor to behave like intrinsic semiconductor for different doping
 - Why does temperature increase to make an extrinsic semiconductor behave like an intrinsic



Q21. Band-diagram change with temperature:

- What all change occurs in a band-diagram with temperature
- See the figure below and observe the band-diagram change with temperature. Also, notice the different carrier concentrations with temperature. MATLAB Codes are shared in resources folder for practise and visualization.
 - Describe the E_f position at $T = 0K$, $\sim 100K$ and $> 600K$
 - If doping is increased to $10^{18} /cm^3$, how much temperature is required for E_f to merge with E_i ?



Q22. Read Section 5.1 from Advanced Semiconductor Fundamentals from Robert Pierret book.

- Describe various recombination and generation processes in 1-2 lines.